As with the situation with superpartners, no such exotically charged particles have ever been observed, and our understanding of string theory does not allow for a definitive prediction of their masses should the extra dimensions have the correct properties to generate them. One explanation for not seeing them, again, is that if they do exist, their masses must be beyond our present technological means—in fact, it is likely that their masses would be on the order of the Planck mass. But should a future experiment come across such exotic electric charges, it would constitute very strong evidence for string theory.

Some Longer Shots

There are yet other ways in which evidence for string theory might be found. For example, Witten has pointed out the long-shot possibility that astronomers might one day see a direct signature of string theory in the data they collect from observing the heavens. As encountered in Chapter 6, the size of a string is typically the Planck length, but strings that are more energetic can grow substantially larger. The energy of the big bang, in fact, would have been high enough to produce a few macroscopically large strings that, through cosmic expansion, might have grown to astronomical scales. We can imagine that now or sometime in the future, a string of this sort might sweep across the night sky, leaving an unmistakable and measurable imprint on data collected by astronomers (such as a small shift in the cosmic microwave background temperature; see Chapter 14). As Witten says, “Although somewhat fanciful, this is my favorite scenario for confirming string theory as nothing would settle the issue quite as dramatically as seeing a string in a telescope.”19

Closer to earth, there are other possible experimental signatures of string theory that have been put forward. Here are five examples. First, in Table 1.1 we noted that we do not know if neutrinos are just very light, or if in fact they are exactly massless. According to the standard model, they are massless, but not for any particularly deep reason. A challenge to string theory is to provide a compelling explanation of present and future neutrino data, especially if experiments ultimately show that neutrinos do have a tiny but nonzero mass. Second, there are certain hypothetical processes that are forbidden by the standard model, but that may be allowed by string theory. Among these are the possible disintegration of the proton (don’t worry, such disintegration,